

MS4 Authority Readiness Status (Filled Assessment)

Assessment date: 2026-04-16
Scope: Current repository state and local verification performed in this workspace

Executive Decision

Current authority decision: **NO-GO**

Reason: critical gates lack operational evidence and real-event performance validation, even though several engineering fixes are now implemented.

Gate-by-Gate Status

Gate	Authority Gate Status	Engineering Implementation Status	Evidence in Repo	Blocking Gap
A. Security & credential hygiene	FAIL	Implemented	<code>scripts/secret_scan.sh</code> , <code>.github/workflows/ci.yml</code> , <code>.env.example</code> , <code>docs/SECURITY_ROTATION.md</code> , <code>.gitignore</code>	Credential rotation completion evidence is missing; no documented proof old keys were revoked.
B. Runtime architecture integrity	FAIL	Implemented	<code>start_local_microservices.ps1</code> , <code>docker-compose.yml</code> , <code>docker-compose-local.yml</code> , <code>api-gateway/src/main/resources/application.yml</code>	Missing runtime proof package (<code>/status</code> , <code>/api/v1/radar</code> , service logs) for local and Docker full-stack runs.
C. Data freshness & degrade behavior	FAIL	Implemented	<code>src/daemon.py</code> (<code>get_chirps_recency_days</code> , <code>precipitation_source</code> , <code>source_health</code> , <code>daemon_status=stale</code>)	Missing formal stale-source test report and captured payload evidence from live run.

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D. Canonical feature schema consistency	FAIL	Implemented	<code>src/feature_schema.py</code> , <code>src/inference_shared.py</code> , <code>src/model.py</code> , <code>src/training_events.py</code> , <code>src/api.py</code>	Need retrained/re-saved production model artifact and integration test report proving all inference paths pass with deployed model.
E. Historical backtest performance	FAIL	Not completed	Checklist thresholds defined in <code>docs/AUTHORITY_READINESS_CHECKLIST.md</code>	No authoritative Tunisia backtest report (including Nabeul/Mahdia rainy periods) with required metrics.
F. Alert reliability & governance	FAIL	Partially implemented	<code>.github/workflows/poll_gee.yml</code> now uses shared inference, hazard type, lat/lon, cap/dedupe	Missing approved operational SOP and controlled dry-run evidence with authority reviewers.
G. Operational hardening & shadow mode	FAIL	Not completed	N/A	No 14-28 day shadow mode, no uptime SLO evidence, no post-incident review record.

Verified Engineering Evidence (Code-Level)

- Secret scanning and CI enforcement:
 - `scripts/secret_scan.sh`
 - `.github/workflows/ci.yml` (secret-scan step)
- Local and Docker reliability wiring:
 - `start_local_microservices.ps1` starts API + daemon + gateway + frontend
 - `docker-compose.yml` and `docker-compose-local.yml` include `disaster-daemon` and shared cache volume
- Gateway local/dev URL routing:
 - `api-gateway/src/main/resources/application.yml` uses `DISASTER_DETECTION_URL`, `DISASTER_DETECTION_WS_URL`, `FRONTEND_URL`
- Canonical schema and shared inference:

- `src/feature_schema.py`
- `src/inference_shared.py`
- `src/model.py` schema metadata checks (`canonical-v1`)
- `src/training_events.py` canonical mapping
- Freshness and stale logic:
 - `src/daemon.py` exposes CHIRPS lag, source health, fallback logic, stale status
- Frontend 5-second live radar polling and status display:
 - `nexus-aid-frontend/src/hooks/useRadar.ts`
 - `nexus-aid-frontend/src/pages/crisis/Dashboard.tsx`

Required Actions to Reach GO

1. Security closure evidence

- Record key revocation timestamps and replacement credentials rollout.
- Attach signed incident closure note.

2. Model validation package

- Run event-based backtest on Tunisia truth set (include Nabeul and Mahdia heavy-rain windows).
- Publish hazard-level precision/recall/FNR/calibration and threshold rationale.

3. Runtime acceptance package

- Capture and archive:
 - local run logs
 - Docker run logs
 - `/status` and `/api/v1/radar` snapshots over multiple daemon cycles
- Include one forced stale-data exercise showing expected behavior.

4. Governance package

- Approve operator SOP for normal/stale/offline modes.
- Run shadow mode (14-28 days) and publish outcomes.

Final Recommendation

You now have a much stronger engineering baseline, but for authority deployment today this remains **NO-GO** until evidence-based validation gates are closed.